



Optimizing MSM registration parameters fixes local minima issues on the dHCP database

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early brain development cortical folding pattern method cortical surface spherical surface registration MSM

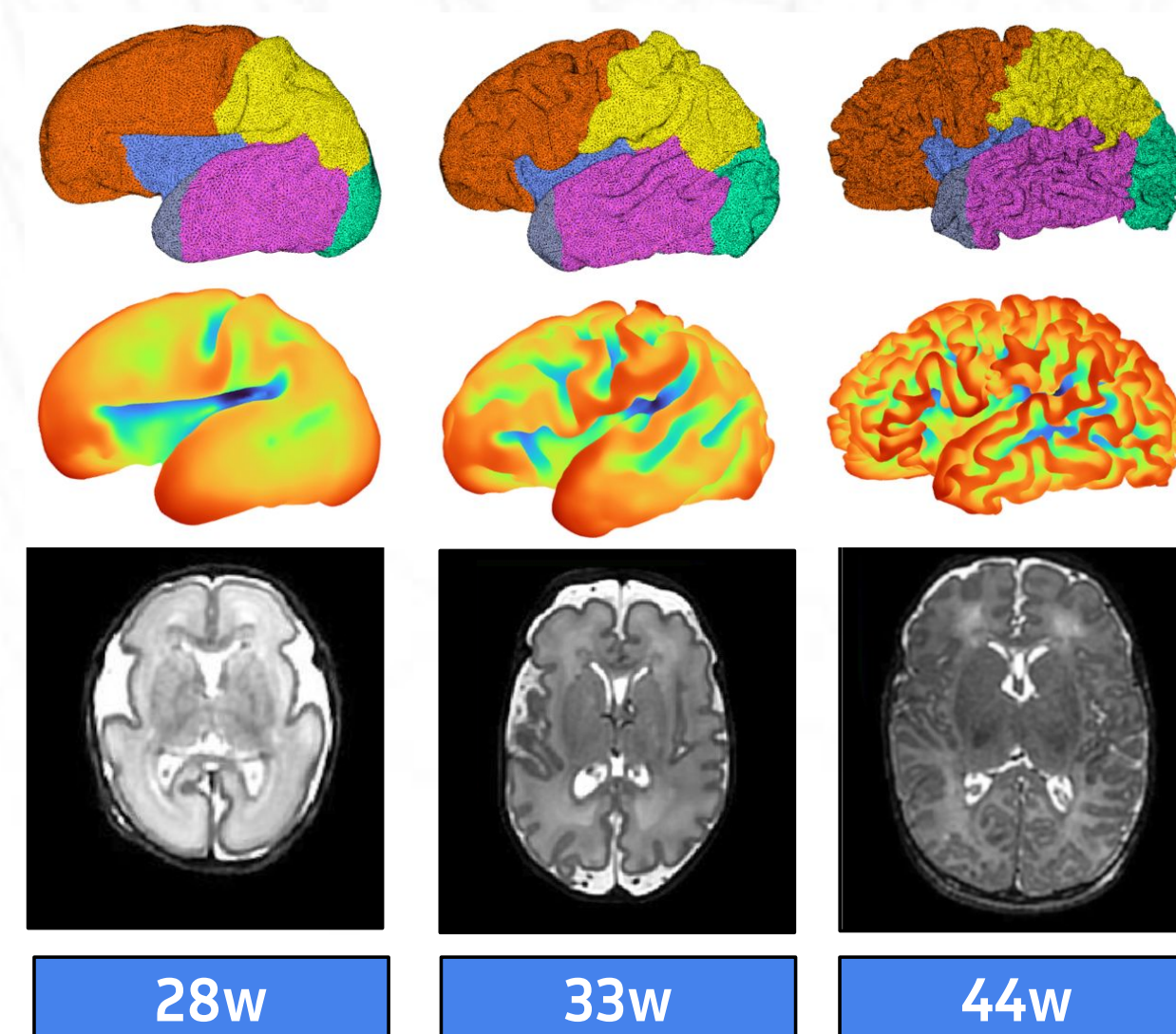
CONTEXT

- **Study of early brain development:** fetal & neonatal MRI-based cortical surface reconstruction
- **Key challenge:** high variability in cortical folding and functional region localization
- **Group studies & statistics:** requiring accurate inter-subject spatial correspondence
- **MSM¹:** state-of-the-art spherical surface registration method for neuroimaging (dHCP pipelines)

DATA

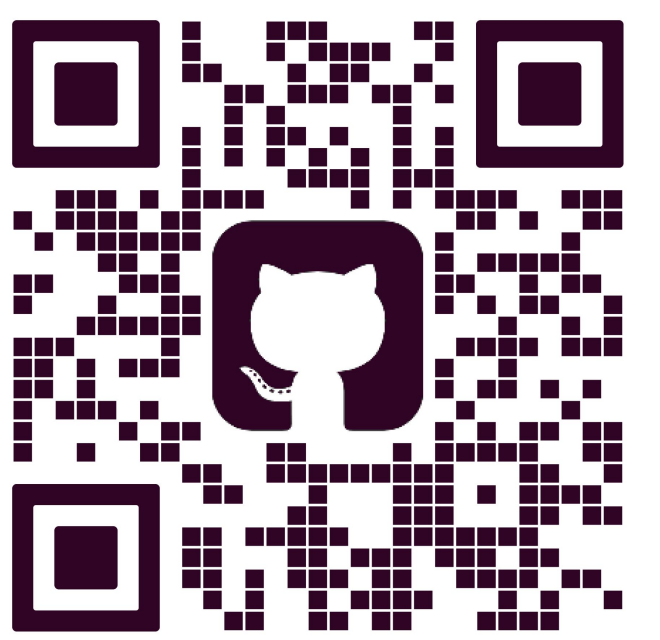
We used the **neonatal** cohort from the dHCP³:

- **Age range:** 28 - 45 postmenstrual weeks
- **Acquisitions (n = 873)⁴:**
 - Right-hemisphere cortical surfaces
 - Sulcal depth features
 - Cortical parcellations (17 regions)
- **Age-matched atlas⁵:**
 - Right-hemisphere cortical surfaces
 - Sulcal depth features



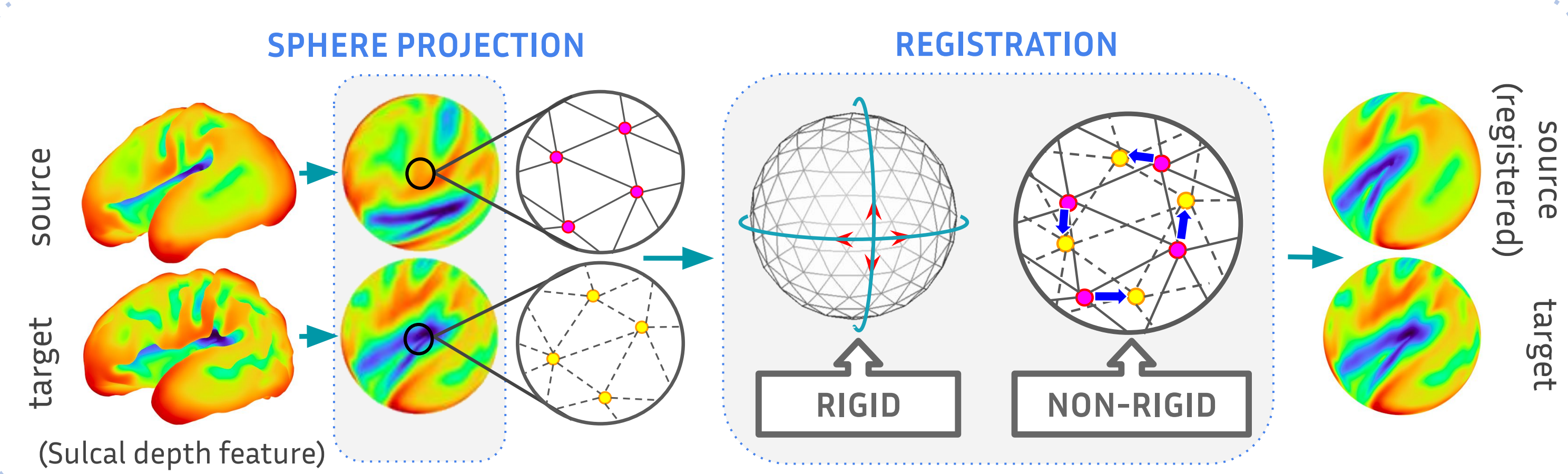
OUR CONTRIBUTION

- 1) **Suboptimal registrations:** quantified impact of default parameters
- 2) **Parameter optimization:** additional rigid step using σ -controlled smoothed sulcal depth features
- 3) **Validation:** quantified improvements in registration accuracy and robustness



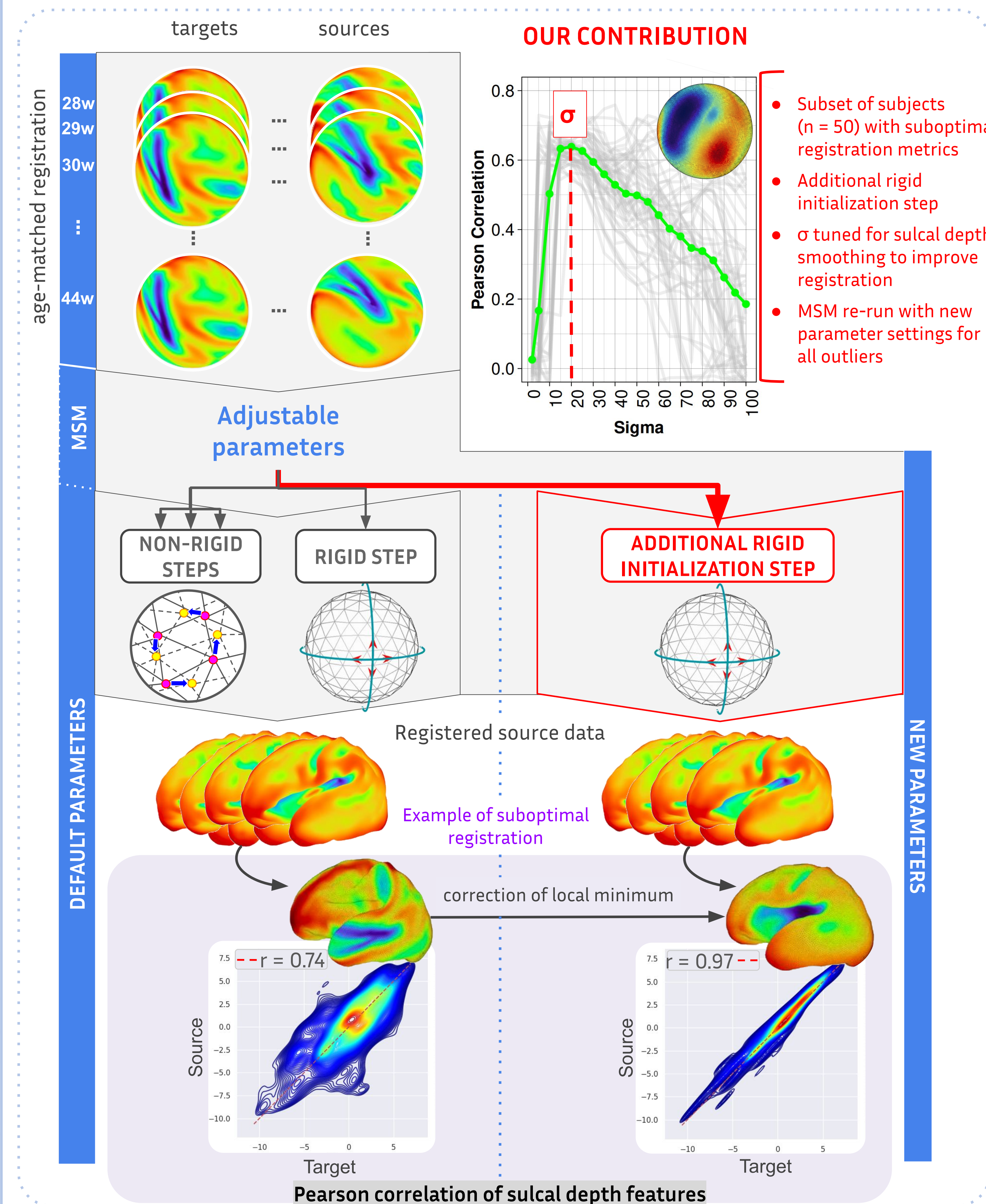
METHODS & RESULTS

PRINCIPLES OF SPHERICAL SURFACE REGISTRATION

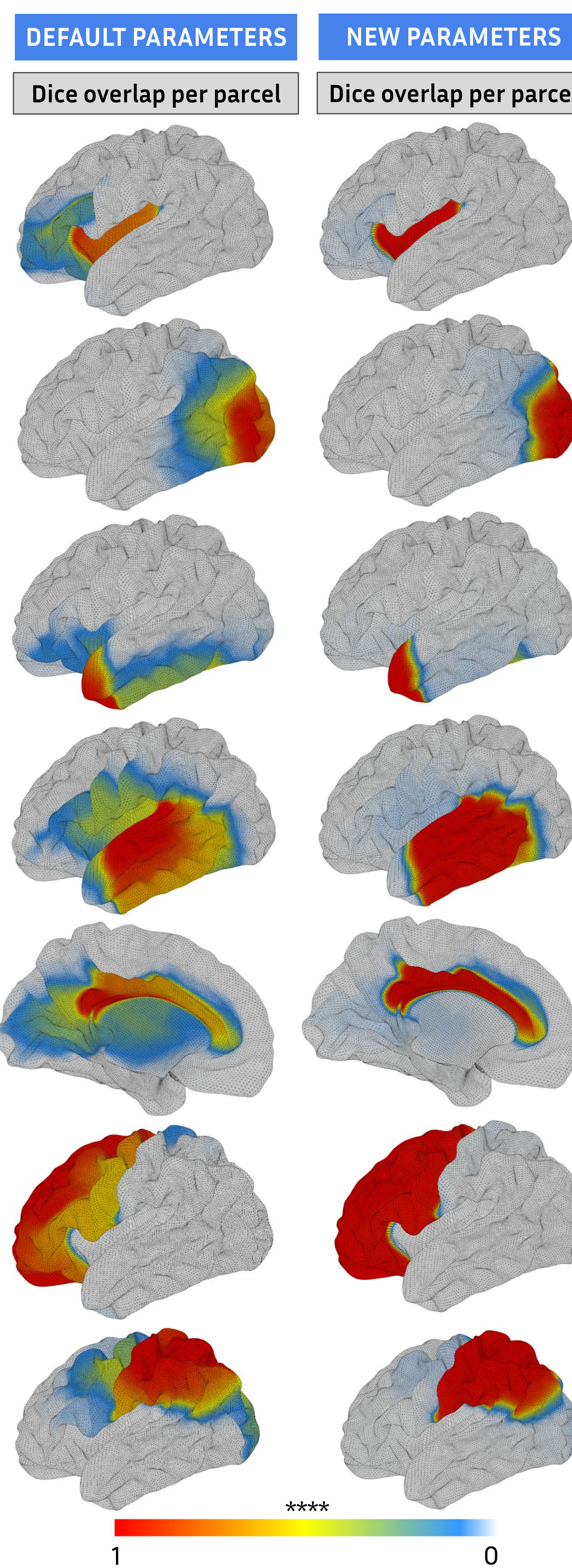


Good registrations: 70%, n=612 Failed registrations: n=211 Optimization cohort: n=50
Age effect: 34-44 PMA

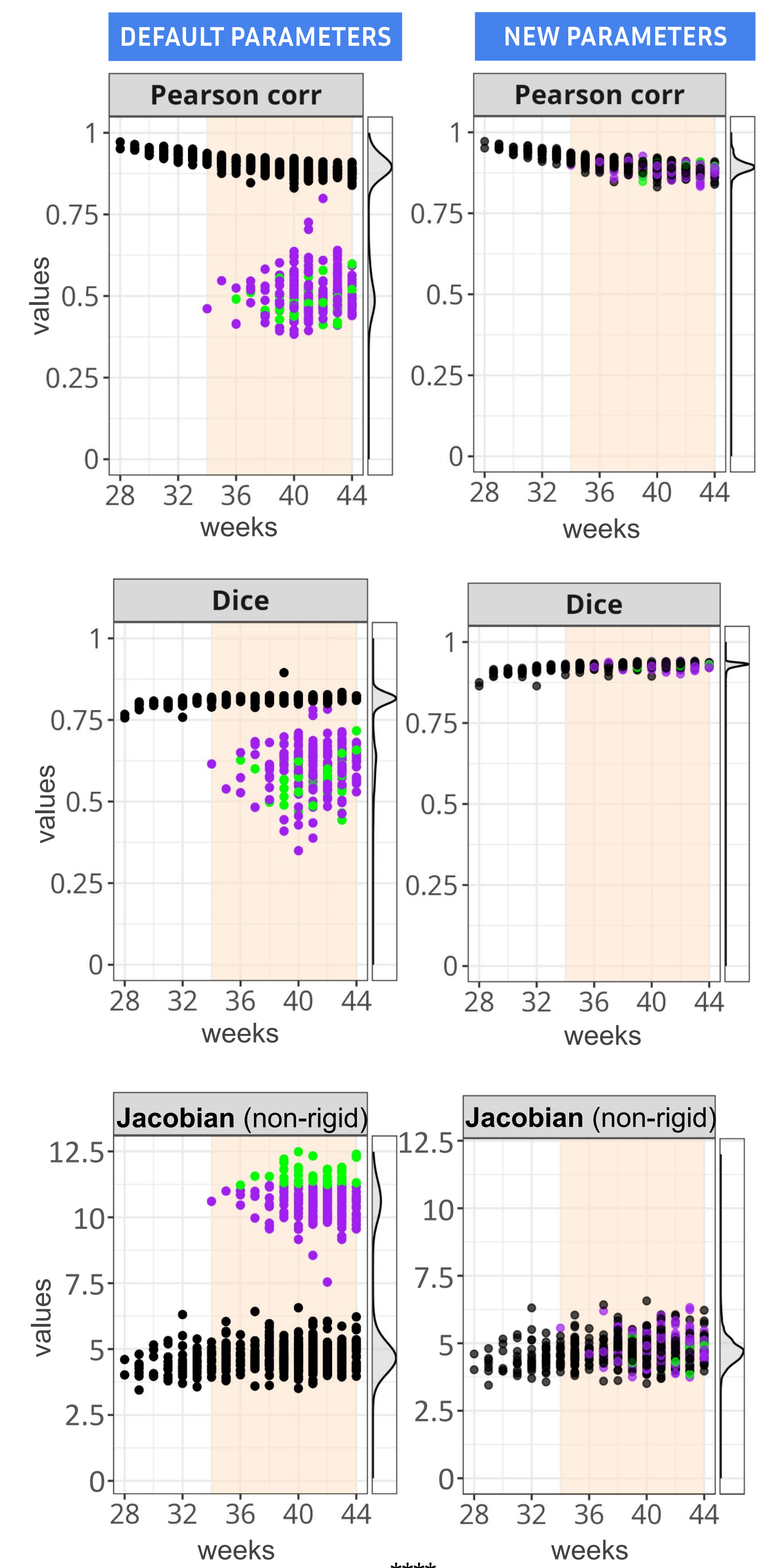
MSM REGISTRATION



QUALITATIVE (across all subjects)



QUANTITATIVE



CONCLUSIONS

- **Suboptimal registrations:** ~30% failure rate in dHCP using default MSM parameters
- **Neonatal studies:** potential impact on developmental trajectories and morphometric analyses
- **Parameter optimization:** consistent improvement across all registration metrics

REFERENCES

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